

Lecture 1 - Introduction

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October 22, 2025

- A *sequential* system is a system composed of a single part which performs actions according to some defined order.
- A *processing element* is defined as an environment in which a process is run, which is capable of allocating resources such as memory to the process running on it.
 - By default, the word “process” refers to the execution of a sequential system.
- *Concurrent* and *parallel* systems are systems composed of independently operating parts functioning together to perform some task.
 - Concurrent and parallel systems are not the same thing, although they are often used interchangeably.
- Concurrent systems consist of processes which perform actions with logical simultaneity.
 - Two concurrently running processes interleave their actions on the same processing element.
- Parallel systems consist of processes which perform actions with physical simultaneity.
 - Two processes running in parallel require that each process run on a separate processing element.
- Concurrent systems can be described as the composition of two or more sequential processes.
- There exist many ways to model concurrent systems:
 1. *Finite State Processes* (FSPs)
 - Finite State Processes are an algebraic description of sequential and concurrent processes.
 - Also called *process algebras*.
 2. *Labelled Transition Systems* (LTSs)
 - LTSs are extensions of finite state machines used to visually represent sequential and concurrent processes.
 - *Labelled Transition System Analyzers* (LTSAAs) are tools used to visualize, animate, and analyze the behaviour of concurrent systems.
 3. *Petrinets*
 - Petrinets are an alternate visualization of concurrent processes.